PRODUCT REQUIREMENT DOCUMENT

(PRD)

UM Hackathon 2026

The Bare Minimums

Domain: AI for Economic Empowerment & Decision Intelligence

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Table of Contents

1 Project Overview

1.1 Problem Statement

Small and medium-sized food and beverage businesses in Malaysia often face operational challenges because most owners still depend on manual record keeping, spreadsheets, or disconnected software tools to manage daily operations. As a result, important decisions such as inventory replenishment, cost control, and profit monitoring are often delayed until problems become severe. Many business owners lack the time and technical expertise needed to analyse their business data regularly, while enterprise-level business intelligence systems remain too expensive and overly complex for smaller operators. This creates a gap where small businesses generate valuable operational data but are unable to convert it into practical decisions that improve performance.

1.2 Target Domain

FnB.ai is designed specifically for the **food and beverage SME market in Malaysia**, including cafés, independent restaurants, bakeries, food stalls, and small restaurant chains. The platform focuses on businesses that require affordable digital transformation tools without needing dedicated IT staff or advanced technical knowledge.

1.3 Proposed Solution Summary

FnB.ai is a cloud-based SaaS platform that uses artificial intelligence to transform business data into actionable operational insights. The platform combines financial tracking, inventory monitoring, anomaly detection, automated reporting, and a conversational AI advisor into a single system. Instead of forcing business owners to manually interpret spreadsheets, the system automatically identifies issues, highlights opportunities, and recommends practical next steps that can improve profitability and efficiency.

2 Background & Business Objective

2.1 Background of the Problem

In many small F&B businesses, owners personally handle multiple responsibilities including procurement, staffing, accounting, and customer service. Because of these demands, financial and operational analysis is often postponed or ignored entirely. This leads to common issues such as over-ordering stock, underestimating expenses, missing profitable trends, and reacting to business problems only after revenue has already been affected. Existing restaurant management software often prioritizes POS functionality, but many platforms fail to provide intelligent business guidance that smaller operators can easily understand and use.

2.2 Importance of Solving This Issue

Solving this problem creates both operational and economic value for small businesses. By reducing the amount of time spent on manual analysis, owners can focus more on customers and strategic growth. Early identification of unusual expenses or declining sales can prevent larger financial losses. Better inventory forecasting can reduce food waste and stock shortages, while AI-generated recommendations can help owners make faster decisions with greater confidence. The ability to provide enterprise-level insight at an SME-friendly cost can significantly improve business sustainability.

2.3 Strategic Fit/ Impact

FnB.ai aligns with the broader movement toward AI-assisted business operations by making advanced analytics accessible to underserved SMEs. Rather than replacing existing workflows entirely, the platform enhances current business practices by embedding intelligence into routine operations. This creates a practical pathway for digital transformation in a market segment that has traditionally been underserved by enterprise software vendors.

3 Product Purpose

3.1 Main Goal of the System

The primary goal of FnB.ai is to provide small F&B business owners with a digital AI assistant that continuously monitors operational performance and delivers useful recommendations. The system is intended to simplify business management by reducing manual effort, improving visibility into business health, and enabling faster decision-making through natural language interaction and automated reporting.

3.2 Intended Users (Target Audience)

Primary Users

The system is primarily intended for:

- Small café owners
- Independent restaurant operators
- Restaurant managers
- Food truck owners
- Bakery owners

These users typically manage daily operations themselves and require a system that is simple enough to use without financial or technical training.

User Characteristics

Most users are expected to:

- Have moderate digital literacy
- Use smartphones and web applications regularly
- Understand basic business operations
- Need quick insights instead of complex reports
- Prefer simple, actionable recommendations over technical analysis

4 System Functionalities

4.1 Description

FnB.ai functions as an intelligent business management platform that gathers information from daily business activities such as sales records, inventory usage, and expenses. This information is processed by the system and transformed into visual dashboards, predictive alerts, and AI-generated insights. The platform is designed to make complex business data understandable for non-technical users while preserving enough detail for meaningful decision-making.

4.2 Key Functionalities

Dashboard Monitoring

The dashboard serves as the central overview of business performance. It displays current revenue, expenses, profit margin, and stock alerts in real time. Visual charts allow users to identify trends quickly, while quick action buttons allow immediate access to common tasks such as importing data or checking AI recommendations. This ensures that users can understand business performance within minutes of logging in.

AI Daily Briefing

Each day, the system automatically generates a personalized business summary that highlights key performance indicators, unusual changes, inventory risks, and recommended actions. This allows users to begin each day with a concise understanding of what requires attention, reducing the need for manual review of reports.

Zara AI Chatbot

The conversational AI assistant enables users to ask natural language questions about their business. Instead of navigating multiple reports, users can ask questions such as why revenue dropped, what items need restocking, or how expenses can be reduced. The chatbot uses business context to produce relevant responses tailored to the user’s current data.

Inventory Management

The inventory module tracks stock quantities, identifies low stock levels, monitors expiry dates, and logs every inventory adjustment. This helps reduce waste while preventing supply shortages that could interrupt daily operations.

Financial Monitoring

The finance module calculates profit and loss, tracks recurring expenses, identifies unusual spending patterns, and estimates burn rate. These features help users understand not just what happened in the business, but why it happened.

4.3 AI Model & Prompt Design

4.3.1 Model Selection

FnB.ai uses **Google Gemma 4 26B through Google AI Studio** as its core language model. The model was selected because it can process business-related context efficiently while generating clear, human-readable recommendations. Its ability to summarize operational data makes it suitable for both automated reporting and interactive business assistance.

4.3.2 Prompting Strategy

The system uses a structured prompting strategy where multiple data sources are assembled before generating a response. Recent sales data, expense records, inventory levels, anomalies, and business rules are combined into a contextual prompt so the AI can produce more personalized recommendations. This approach improves response quality while reducing generic outputs.

4.3.3 Context & Input Handling

To prevent excessive token usage, the system summarizes raw business data before sending it to the model. Large datasets are converted into concise snapshots containing the most relevant information. This ensures the AI receives enough business context without unnecessary processing overhead.

4.3.4 Fallback & Failure Behaviour

If the AI generates a poor or irrelevant response, the system can retry generation using adjusted prompts. Cached summaries may also be shown temporarily if the AI service becomes unavailable. When failures occur, users receive simple and understandable messages rather than technical errors, preserving trust in the platform.

5 User Stories & Use Cases

5.1 User Stories

Business Owner

As a business owner, I want to ask questions in natural language so that I can get insights without reading complex reports.

Inventory Manager

As a restaurant manager, I want to know which inventory items are running low so that I can restock before service is affected.

Financial Monitoring

As a financial monitor, I want to receive a daily AI-generated summary so that I can quickly understand my business performance before opening for the day.

5.2 Main Use Cases

Daily Briefing Workflow

The user logs into the platform, the system retrieves recent operational data, the AI generates a briefing, and the user reviews recommended actions before starting daily operations.

Inventory Alert Workflow

When stock falls below a predefined threshold, the system generates an alert on the dashboard. The user can then open the inventory page and take corrective action immediately.

6 Features Included (Scope Definition)

The MVP includes:

- User authentication
- Business onboarding
- Dashboard analytics
- AI daily briefings
- Zara AI chatbot
- Inventory management
- Financial monitoring
- Business rules customization
- CSV sales data import
- Demo data seeding

These features represent the minimum complete workflow required to demonstrate the value of AI-assisted business management for F&B SMEs.

7 Features Not Included (Scope Control)

To maintain a manageable MVP scope, several advanced features are intentionally excluded from the current version. These are planned for future versions.

The current version does not include:

- Multi-user staff permissions
- Mobile application
- OCR invoice scanning
- POS direct integrations
- Supplier marketplace
- Multi-location management
- Public API access
- WhatsApp integration

8 Assumptions & Constraints

LLM Cost Constraint

To control AI usage cost:

- AI briefings are cached for 24 hours
- Repeated requests avoid regeneration
- Context is summarized before sending
- Free-tier AI limits are considered during development
- A typical 30-minute user session is estimated to consume approximately **19,400 total tokens** (13,500 input tokens and 5,900 output tokens), allowing prompt design to be monitored and optimized for long-term cost efficiency.

Technical Constraints

The project currently depends on:

- Next.js frontend
- Supabase backend
- Google AI Studio
- CSV-based imports instead of live POS integrations

Performance Constraints

Target performance:

- Dashboard under 2 seconds
- Chat responses under 5 seconds
- AI briefing under 10 seconds

These targets ensure the system remains practical during daily business usage.

User Input Assumption

It is assumed users:

- can upload CSV files
- can manually enter inventory
- have internet access
- understand basic digital interfaces

9. Risks & Questions Throughout Development

AI Reliability Risk

AI-generated recommendations may occasionally produce inaccurate, incomplete, or overly generalized advice if the model misinterprets business data. Because the platform provides operational and financial suggestions, unreliable outputs could reduce user confidence in the system. To minimize this risk, the platform validates source data before analysis, applies structured prompting techniques, and allows users to regenerate responses when clarification is needed. Continuous refinement of prompts and response monitoring will also be necessary throughout development to improve recommendation accuracy.

Adoption Risk

Some business owners may hesitate to trust AI-driven tools or may be reluctant to replace familiar manual workflows with a digital system. Resistance to adoption could limit the practical impact of the platform even if the technology performs well. To address this challenge, the onboarding process includes demo data, guided walkthroughs, and simple explanations that clearly show the immediate business value of the platform. The system is also designed to present recommendations in a straightforward and understandable way to reduce the learning barrier for first-time users.

Scalability Risk

As the number of businesses using the platform increases, larger datasets and more frequent AI requests may affect response speed and system stability. Delays in dashboard loading or chatbot performance could reduce overall usability. This risk is reduced through serverless deployment, response caching, optimized database queries, and efficient context preparation before AI processing. These measures help ensure that performance remains stable as the platform scales.

Financial Risk

Cloud infrastructure and AI processing costs may increase as user activity grows, especially when generating daily briefings and chatbot responses at scale. If unmanaged, rising operational costs could affect the long-term sustainability of the platform. To reduce this risk, the system emphasizes cost-efficient AI usage through caching, summarized context handling, and controlled API requests.

A future subscription-based pricing model is also planned to support sustainable growth while maintaining affordability for small businesses.